

VADEROARNA, Sweden

WMO: 025010

Lat: 58.576N

Lon: 11.066E

Elev: 79

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.7	18.5	8.3	8.5	17.4	11.7	10.0	22.3	43.0	39.2	39.8	39.3	16.2	60	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	73.2	64.2	70.4	63.6	68.2	62.6	66.5	70.9	65.0	68.8	63.5	67.0	11.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.7	68.7	63.3	87.3	67.3	61.9	83.1	65.9	31.2	71.0	30.0	68.8	28.9	67.1	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.7	36.7	33.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	33.1	31.9	36.3	43.4	52.4	58.9	63.4	62.5	57.6	49.4	42.3	36.9
	DBStd	12.28	6.88	6.73	5.00	4.72	5.80	4.11	3.85	2.87	3.61	4.43	6.30	7.91
	HDD50	2403	523	508	425	206	35	0	0	0	2	64	233	407
	HDD65	6464	988	928	890	647	394	191	78	90	223	483	680	872
	CDD50	1464	0	0	0	9	109	266	416	386	229	46	3	0
	CDD65	49	0	0	0	0	2	7	29	11	0	0	0	0
	CDH74	53	0	0	0	0	4	9	39	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	17.3	18.7	17.3	16.7	15.0	15.2	15.1	14.9	16.3	18.2	19.7	20.7	20.2
(15) Precipitation	PrecAvg	34.0	3.0	2.2	1.7	2.0	2.1	3.1	2.9	3.4	3.1	4.0	3.5	3.0
	PrecMax	48.3	6.4	5.9	4.7	4.3	3.9	6.5	6.1	6.2	7.4	9.6	9.2	6.6
	PrecMin	23.0	0.4	0.6	0.1	0.5	0.0	0.2	0.4	0.7	1.2	0.8	1.1	0.6
	PrecStd	5.9	1.5	1.4	1.0	1.0	1.0	1.3	1.3	1.3	1.5	2.0	1.9	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.6	43.2	48.2	61.2	73.0	74.7	77.6	72.8	67.2	59.2	54.6	50.3
		MCWB	42.5	41.1	43.5	50.8	61.1	61.3	66.0	64.6	61.5	56.0	54.4	48.6
	2%	DB	44.2	42.1	45.1	55.7	68.1	70.5	74.8	70.0	64.7	57.8	52.2	48.5
		MCWB	41.8	40.3	42.1	48.7	58.9	61.7	65.9	64.1	60.6	55.4	50.8	46.6
	5%	DB	42.9	41.2	43.4	52.4	64.3	67.2	71.7	68.4	63.4	56.9	51.1	47.1
		MCWB	40.9	39.8	40.8	46.9	56.9	60.6	65.4	63.3	59.5	54.6	49.7	45.1
	10%	DB	41.8	40.2	42.1	50.0	60.8	64.8	69.1	66.9	62.2	55.8	49.7	45.7
		MCWB	39.6	38.9	39.8	45.8	55.2	58.8	63.7	62.0	58.5	53.2	48.5	43.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	42.0	44.3	52.9	62.5	65.5	68.6	67.3	63.7	57.6	54.7	49.2
		MCDB	44.4	42.9	46.7	58.9	71.0	69.3	74.3	69.9	65.4	58.0	54.7	50.0
	2%	WB	42.2	41.1	42.9	50.2	60.0	63.1	67.2	65.7	62.4	56.6	51.6	47.7
		MCDB	43.1	41.8	44.6	54.5	65.9	67.4	72.5	68.8	63.9	57.3	51.8	48.4
	5%	WB	41.1	40.3	41.7	48.2	58.2	61.5	65.9	64.2	60.9	55.1	50.2	46.0
		MCDB	42.4	41.1	43.3	52.4	63.1	66.0	70.6	67.0	62.7	56.3	50.9	47.0
	10%	WB	39.9	39.3	40.5	46.2	56.2	59.9	64.5	62.8	59.6	53.6	48.7	44.4
		MCDB	41.4	40.4	41.9	49.6	60.3	64.1	68.4	65.8	61.7	55.1	49.7	45.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	4.8	5.0	6.3	6.7	5.9	5.5	5.0	5.0	5.1	4.5	5.3
		MCDBR	6.1	4.2	5.7	9.7	11.2	9.4	9.1	7.2	5.7	4.8	5.0	4.6
	5% WB	MCWBR	6.4	4.3	4.3	6.0	6.6	6.3	5.5	5.6	4.9	4.6	5.4	5.8
		MCWBR	5.8	4.3	5.3	9.1	10.1	7.8	8.0	5.8	4.7	4.1	4.8	4.5
(40) Clear-Sky Solar Irradiance	taub		0.294	0.326	0.343	0.381	0.375	0.363	0.385	0.383	0.353	0.340	0.310	0.274
		taud	2.243	2.317	2.360	2.308	2.408	2.468	2.427	2.427	2.493	2.417	2.286	2.178
	Ebn at Noon		168	221	253	261	272	277	267	259	250	214	160	133
		Edh at Noon	15	22	28	35	33	32	33	30	24	20	14	11
(44) All-Sky Solar Radiation	RadAvg		105	285	758	1260	1702	1924	1760	1372	910	407	138	70
	RadStd		19	54	73	164	149	110	174	84	90	59	30	7

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page